

INDEX

Chapter No.	Title	Page No.
1	Introduction	
	1.1. Introduction to the project	1
	1.2. Motivation for the project	1
	1.3. Drawbacks of the existing system	1
	1.4. Problem Definition	2
	1.5 Relevance of the Project	2
	1.6 Methodology used	3
2.	Literature Survey 2.1. Research Papers 2.2 Software Referred / Article Referred	4
3.	Requirement Of Proposed System 3.1 Functional Requirements 3.2. Non-Functional Requirements	8

	3.3. Constraints 3.4. Hardware & Software Requirements 3.5. Techniques utilized 3.6. Tools utilized 3.7. Project Proposal	
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4.	Proposed Design 4.1 Block diagram representation of the proposed system: Explanation for the block diagram 4.2. Modular diagram representation of the proposed system. Explanation for the modular block diagram 4.3 Design of the proposed system with proper explanation of each : a. Data Flow Diagrams b. ER Diagram c. Screenshot of implementation 4.4 Algorithms utilized in the existing systems	12
5.	Proposed Results and Discussions 5.1.Determination of efficiency 5.2. Determination of accuracy 5.3. Reports on sensitivity analysis 5.4. Graphs of : Accuracy Vs time	20
6.	Plan Of Action For the Next Semester 6.1. Work done till date 6.2.Plan of action for project II	28
7.	Conclusion	30
8.	References (In IEEE format)	31
9.	Appendix 9.1. List Of Figures 9.2. List Of Tables	33

Introduction

1.1 Introduction

Maternal health remains one of the most critical global health challenges, especially in low-income regions where access to quality prenatal care is limited. The World Health Organization (WHO) recommends that every pregnant woman receive at least one ultrasound scan before 24 weeks of gestation. Such screening is essential for accurately determining **gestational age (GA)**—a key indicator for monitoring fetal growth and identifying high-risk pregnancies. Accurate estimation of GA allows timely interventions and significantly reduces preventable maternal and neonatal complications.

Natalis is a proposed smartphone-based AI-assisted ultrasound system designed to estimate **gestational age** from standard ultrasound images or videos. The system empowers minimally trained health workers to perform basic obstetric screening using affordable, portable ultrasound probes. By analyzing only visual data, Natalis automates GA measurement and provides clear, interpretable results directly on a smartphone. Designed to function offline with minimal power requirements, Natalis aims to deliver an accessible, low-cost, and reliable prenatal screening solution for underserved and resource-limited settings.

1.2 Motivation for the Project

Despite significant medical advancements, approximately **295,000 women** die annually from preventable pregnancy-related complications, with **99% of these deaths** occurring in low- and middle-income countries. A major contributing factor is the lack of access to timely ultrasound screening, primarily due to the scarcity of trained sonographers and functional imaging equipment in rural clinics.

Recent innovations in AI and portable ultrasound technology demonstrate that diagnostic imaging can be democratized. Systems like **BabyChecker** and **DeepEcho** have proven that artificial intelligence can guide untrained health workers to acquire diagnostic-quality scans and interpret them accurately. Motivated by these breakthroughs, Natalis seeks to extend such capabilities to a wider audience through an intuitive Android application capable of real-time, on-device image analysis—empowering frontline workers and improving maternal health outcomes globally.

1.3 Drawbacks of the Existing System

Existing obstetric ultrasound systems face several limitations in low-resource settings:

- **Dependence on Expert Operators:** Traditional systems require trained sonographers, limiting use in remote areas.
- **High Equipment Costs:** Conventional ultrasound machines are expensive and require regular maintenance.

- **Power and Connectivity Constraints:** Many rural clinics lack reliable electricity or internet, making standard equipment unsuitable.
- **Complex Interfaces:** Most ultrasound systems are designed for clinical specialists, not community health workers.
- **Limited AI Integration:** Current low-cost portable devices often lack automated interpretation or guidance features, making them difficult to use effectively without expertise.

These constraints leave a large portion of the global population without access to essential prenatal screening.

1.4 Problem Definition

The key problem addressed by this project is the **lack of accessible, automated, and low-cost prenatal ultrasound screening** in resource-poor regions. Many expectant mothers never receive an ultrasound due to equipment unavailability or insufficient operator training.

Problem Statement:

To develop a **smartphone-based AI tool—Natalis**—capable of analyzing ultrasound images and videos to automatically estimate gestational age, detect fetal presentation, identify placental location, and count fetuses. The system must operate **offline**, require **minimal user training**, and function efficiently on low-power Android devices, thus enabling early risk detection and referral in underserved communities.

1.5 Relevance of the Project

The Natalis project is highly relevant to the goals of **global maternal health, AI in healthcare, and digital health equity**. By leveraging deep learning to automate image interpretation, Natalis can serve as a **scalable solution** for antenatal screening in rural and remote areas.

Its relevance extends to:

- **Public Health:** Contributes to the reduction of maternal and neonatal mortality.
- **Technology for Social Impact:** Demonstrates the practical use of AI in addressing real-world healthcare disparities.
- **Sustainability:** Operates offline with minimal infrastructure, aligning with the WHO's global digital health strategies.
- **Research Advancement:** Provides a data-driven framework for further exploration of AI-assisted ultrasound diagnostics.

Thus, Natalis bridges the gap between advanced medical imaging and community-level healthcare delivery.

1.6 Methodology Used

The project employs a data-driven deep learning methodology for ultrasound image analysis, combined with a mobile deployment framework for real-world usability. The development process includes the following stages:

1. **Data Collection and Preprocessing:** Ultrasound images and videos are sourced from public datasets and clinical collaborations. Data is preprocessed through denoising, normalization, and augmentation to improve model generalization.
2. **Model Development:** Convolutional Neural Networks (CNNs) and segmentation architectures (such as U-Net or DeepLab) are trained to perform multiple tasks—fetal biometric measurement, presentation classification, placental segmentation, and multi-fetus detection.
3. **System Integration:** The trained model is converted for on-device execution using frameworks like TensorFlow Lite or PyTorch Mobile. The model is integrated into the Natalis Android app, which features a simple user interface designed for health workers.
4. **Offline Inference:** The system performs AI-based analysis directly on the device without internet access, ensuring privacy and functionality in rural environments.
5. **Evaluation and Validation:** The AI outputs are compared against expert sonographer annotations using metrics such as accuracy, Dice coefficient (for segmentation), and mean absolute error (for gestational age estimation).
6. **Deployment and Testing:** Field testing is conducted to evaluate usability, robustness, and real-world diagnostic performance under varying environmental and hardware conditions.

Literature Survey

2.1 Research Papers

Table 1. Literature Survey of Research Papers

Sr No	Title of Paper	Journal/ Conference Date	Abstract	Inference Drawn
1	Estimating Fetal Gestational Age from Ultrasound Images Using Deep Learning Techniques	January 20–22, 2025	This study introduces the UI-DLT framework, a CNN-based system for automatically estimating fetal gestational age (GA) from ultrasound images. It reduces human error, operator dependency, and variability due to image quality or fetal position, achieving over 93% accuracy and 98% success rate.	Deep learning, specifically CNNs, can accurately and consistently estimate gestational age from ultrasound images, minimizing human dependency and enhancing reliability across clinical settings. The approach supports automation and scalability for low-resource healthcare environments.
	Hierarchical Attentive Network for Gestational Age Estimation in Low-Resource Settings	May 15-17, 2023	This study proposes a hierarchical deep sequence learning model with an attention mechanism to estimate gestational age (GA) using low-cost, handheld Doppler ultrasound signals. Data from 226 pregnancies in Guatemala were used, achieving an average error of only 0.79 months, near the theoretical minimum. The model highlights key fetal cardiac activity patterns to identify developmental stages and potential fetal growth restriction (FGR).	The attention-based hierarchical model effectively learns fetal cardiac dynamics for GA estimation using affordable Doppler devices. It shows strong potential for accurate, low-cost, and scalable GA estimation in low-resource settings, minimizing dependence on image quality or operator expertise.

Employing Support Vector Machine Regression to Estimate the Fetal Gestational Age	September 4–7, 2022	This study applies Support Vector Machine (SVM) regression using maternal and fetal heart rate variability (HRV) features extracted from 1-minute ECG recordings of 60 pregnant women. The linear kernel outperformed the RBF kernel, achieving an error of 5.11 weeks and a Pearson correlation of 0.63 with gold- standard GA (CRL). Results suggest maternal HRV plays a vital role in fetal age estimation.	Combining maternal and fetal HRV features in an SVM- linear model enhances GA estimation accuracy, emphasizing maternal cardiac influence on fetal development. Though promising, larger datasets and diverse conditions are needed to improve robustness and generalizability.
Fetal Head and Femur Detection from USG Image to Estimate Gestational Age	October 29–30, 2018	This study proposes an automated approach for detecting and segmenting fetal head and femur in ultrasound (USG) images using Integral Projection. Biparietal Diameter (BPD) and Femur Length (FL) were extracted as key parameters for gestational age (GA) estimation. The method achieved a 97.77% accuracy rate, improving consistency over manual observation.	Preprocessing reduced image noise, and segmentation using Integral Projection accurately identified fetal head and femur. Using BPD and FL as features provided high GA estimation accuracy. Future improvements could include additional parameters like abdominal circumference for enhanced precision.
Automatic Gestational Age Estimation by Femur Length Using Integral Projection from Fetal Ultrasonography	2018	This study presents an automated system for gestational age (GA) estimation using femur length extracted from fetal ultrasound images. Techniques like Gaussian Filtering, Morphological Operations, and Canny Edge Detection were applied for noise reduction and contour detection. Femur length was	The proposed automated method shows strong correlation with manual doctor measurements, minimizing human error and improving diagnostic reliability. It effectively measures femur length and estimates GA with high accuracy, demonstrating potential for clinical use in reducing observer variability.

2.2. Software referred / Articles referred:

Baby Checker

Abstract

Baby Checker is an AI-powered smartphone-based ultrasound solution designed to address the critical global challenge of maternal mortality, particularly in low-resource settings. The system combines artificial intelligence with handheld ultrasound technology to enable frontline healthcare workers to perform pregnancy risk screening with minimal training. BabyChecker analyzes obstetric ultrasound scans to identify risks in pregnancy through a standardized six-sweep protocol across the abdomen, providing essential information including gestational age, fetal presentation, multiple gestations, and placenta location within 2 minutes.

The solution addresses the fact that approximately 800 women die daily from preventable pregnancy-related complications, with 99% of maternal deaths occurring in low-income countries where access to trained sonographers and ultrasound equipment is severely limited. BabyChecker operates offline on Android devices and requires only a 3-minute training video for healthcare workers to become proficient, making it accessible in areas with limited infrastructure and electricity.

Inference

Baby Checker has demonstrated significant clinical impact across multiple deployment sites. Field data shows that 76.3% of first-time users can capture usable scans immediately, indicating high usability from the initial attempt. The system has been successfully implemented in over twelve countries including Sierra Leone, Zambia, Honduras, Kenya, Malawi, and Tanzania, with validation studies confirming its effectiveness in real-world clinical settings.

The AI algorithms powering Baby Checker have been validated to perform pregnancy risk screening that enables early identification and management of high-risk pregnancies at local healthcare facilities. This technology facilitates task-shifting toward frontline health workers, empowering them to screen risky pregnancies more effectively than traditional approaches. The system's deterministic nature ensures consistent outputs, unlike human operators who may have variable performance.

Baby Checker functions as what has been described as a "great equalizer" by the Gates Foundation, enabling any frontline healthcare worker with no prior ultrasound experience to perform essential pregnancy screenings after minimal training. The solution has been recognized for its potential to bridge healthcare disparities and improve maternal health outcomes globally, particularly in underserved regions where traditional ultrasound services are unavailable.

Deep Echo

Abstract

DeepEcho is a cloud-based, standalone software as a medical device (SaMD) that utilizes deep learning algorithms to automate fetal biometry and amniotic fluid volume (AFV) assessments in fetal ultrasound screening. The system is designed to assist qualified healthcare professionals in the analysis and interpretation of fetal ultrasound images for patients aged 18 years or older between the 14th and 41st weeks of pregnancy. Deep Echo processes both DICOM images and live ultrasound streaming to identify standard biometry views (cephalic, abdominal, and femoral) and automatically compute critical measurements, including head circumference (HC), biparietal diameter (BPD), abdominal circumference (AC), femur length (FL), gestational age (GA), and estimated fetal weight (EFW).

The development of Deep Echo was motivated by the need to democratize access to quality fetal ultrasound, particularly addressing the high maternal and fetal mortality rates in regions with limited access to skilled sonographers. The system was trained using one of the world's largest fetal ultrasound annotated datasets, consisting of 172,293 de-identified Moroccan fetal ultrasound images supplemented with publicly available datasets.

Inference

Deep Echo has achieved strong clinical validation through rigorous prospective testing. In a landmark Nature Communications study, its 95% limits of agreement with experts were narrower than intra- and inter-observer variability among human sonographers across all parameters: $\pm 0.54\%$ (HC), $\pm 3.74\%$ (BPD), $\pm 0.14\%$ (AC), $\pm 3.11\%$ (FL), $\pm 1.45\%$ (GA), $\pm 2.42\%$ (EFW), and $\pm 16.96\%$ (SDP). Reliability was excellent, with ICC > 0.9 for all metrics except SDP (ICC = 0.692).

It outperformed state-of-the-art models like FUVAI in mean absolute error for most parameters, and achieved 86.6% sensitivity and 85.7% specificity for polyhydramnios detection. FDA 510(k) clearance in 2025 confirmed its safety and effectiveness. The deterministic design ensures consistent outputs, minimizing variability seen in human measurements.

Fetal ultrasound can lower fetal mortality by up to 20% and maternal mortality by 6%, underscoring the value of democratizing access. Deep Echo's automation enables minimally trained workers to deliver accurate prenatal assessments in resource-limited areas, addressing skilled sonographer shortages and expanding maternal healthcare access.

Requirements of the Proposed System

3.1. Functional Requirements:

1. User Interaction and Input

- The system shall allow users to upload **pre-recorded ultrasound images or short videos** from local storage.
- The system shall provide an **intuitive Android interface** suitable for non-technical users like midwives or field health workers.

2. AI-Based Image Analysis

- The system shall perform **frame extraction** from uploaded videos.
- The system shall identify **standard fetal planes** (head, abdomen, femur) using a **CNN classifier (ResNet)**.
- The system shall **segment fetal structures** to measure biometric parameters (HC, AC, FL) using a **U-Net**.
- The system shall detect **fetal orientation and count** (cephalic/breech, single/twins) using **YOLOv5**.
- The system shall segment the **placental region** and identify its location (anterior/posterior/low-lying).

3. Report Generation

- The system shall compute **gestational age (GA)** based on biometric parameters.
- The system shall display results including:
 - Gestational Age
 - Fetal Presentation
 - Placental Location
 - Fetus Count
- The system shall generate **color-coded risk alerts** and save reports for local record-keeping.

4. Offline Operation

- The system shall function **entirely offline** without internet connectivity.
- The AI inference shall run locally on the smartphone using **TensorFlow Lite / PyTorch Mobile**.

5. Usability & Accessibility

- The interface shall support **multiple languages** and **icon-based visualization** for easy understanding.
- The system shall provide a **quick setup (<3 minutes)** for non-expert health workers.

3.2. Non-Functional Requirements:

1. Performance

- On-device inference shall complete within **2 minutes per scan**.
- The app shall maintain **at least 20 FPS** during video frame extraction.

2. Accuracy

- **GA estimation error $\leq \pm 2$ weeks**
- **Fetal presentation detection >95% accuracy**
- **Placenta segmentation Dice score ≥ 0.90**

3. Reliability

- The app shall handle corrupted or low-quality images gracefully with warning messages.
- The system shall maintain **data consistency** even if the app is interrupted mid-inference.

4. Scalability

- The modular model architecture shall allow **easy integration** of future AI modules (e.g., IUGR, preeclampsia detection).

5. Usability

- Designed for **minimal training requirements** and intuitive navigation.
- Interface shall be **touch-optimized** for Android screens ≥ 5 inches.

6. Security

- All user data (images, reports) shall remain **stored locally** to maintain patient confidentiality.
- No cloud transmission of sensitive data unless explicitly permitted.

3.3. Constraints:

1. The system must **operate offline**, making cloud-based inference unavailable.
2. The mobile device must have **≥ 3 GB RAM** for smooth model inference.
3. AI models must be **lightweight** (optimized for TensorFlow Lite / PyTorch Mobile).
4. The solution must be **cost-effective** and **deployable on entry-level Android phones**.
5. Limited availability of labeled ultrasound datasets may restrict model generalization.
6. Models must function solely on **2D ultrasound images/videos** without metadata or probe tracking.

3.4. Hardware and Software Requirements:

Table 2. Hardware and Software Requirements by Categories

Category	Requirement
Hardware (Training Phase)	GPU-enabled workstation (NVIDIA RTX 3060 or higher, 16GB RAM, 500GB SSD)
Hardware (Deployment Phase)	Android Smartphone (Android 8+, ≥ 3 GB RAM, GPU acceleration)
Software (Model Development)	Python, PyTorch, TensorFlow/Keras, OpenCV, NumPy, Pandas
Software (Mobile Development)	Android Studio (Java/Kotlin), TensorFlow Lite Converter / PyTorch Mobile
Version Control	Git, GitHub
Annotation Tools	LabelImg, CVAT, or custom labeling GUI
Visualization Tools	TensorBoard, Matplotlib
Operating System	Windows / Ubuntu for development; Android for deployment

3.5. Techniques Utilized Till Date for the Proposed System:

Data Preprocessing Techniques

- Frame extraction from ultrasound videos.
- Image normalization (contrast, brightness, denoising).
- Data augmentation (rotation, flip, Gaussian noise) for model robustness.

Deep Learning Techniques

- Convolutional Neural Networks (ResNet): For frame classification and standard plane detection.
- U-Net Architecture: For segmentation of fetal structures and placenta.
- YOLOv5: For fetal orientation and multiple fetus detection.
- Transfer Learning: For reducing training time and improving accuracy on limited datasets

3.6. Tools utilised for Natalis

Table 3. Tools and Libraries used

Category	Tool/Library	Purpose
Development	Streamlit (Python)	Basic UI creation
AI Frameworks	Unet, Resnet	Deep learning model training
Image Processing	OpenCV, PIL	Frame extraction and enhancement
Version Control	Git, GitHub	Source code management
Deployment	Streamlit	Inference on the web-app
Documentation	Google Docs	Report and technical documentation

3.7 Project Proposal

This project proposes **Natalis**, an AI-assisted smartphone-based ultrasound system designed to estimate gestational age from standard fetal ultrasound images or videos. The system combines **computer vision and deep learning** to automatically analyze ultrasound data and provide accurate, real-time gestational assessments without requiring expert operators. By integrating affordable portable probes with on-device AI models, Natalis aims to deliver a **low-cost, offline, and user-friendly prenatal screening solution** for underserved regions, improving early detection of high-risk pregnancies and reducing maternal and neonatal complications.

Proposed Design

4.1 Block diagram representation of Natalis

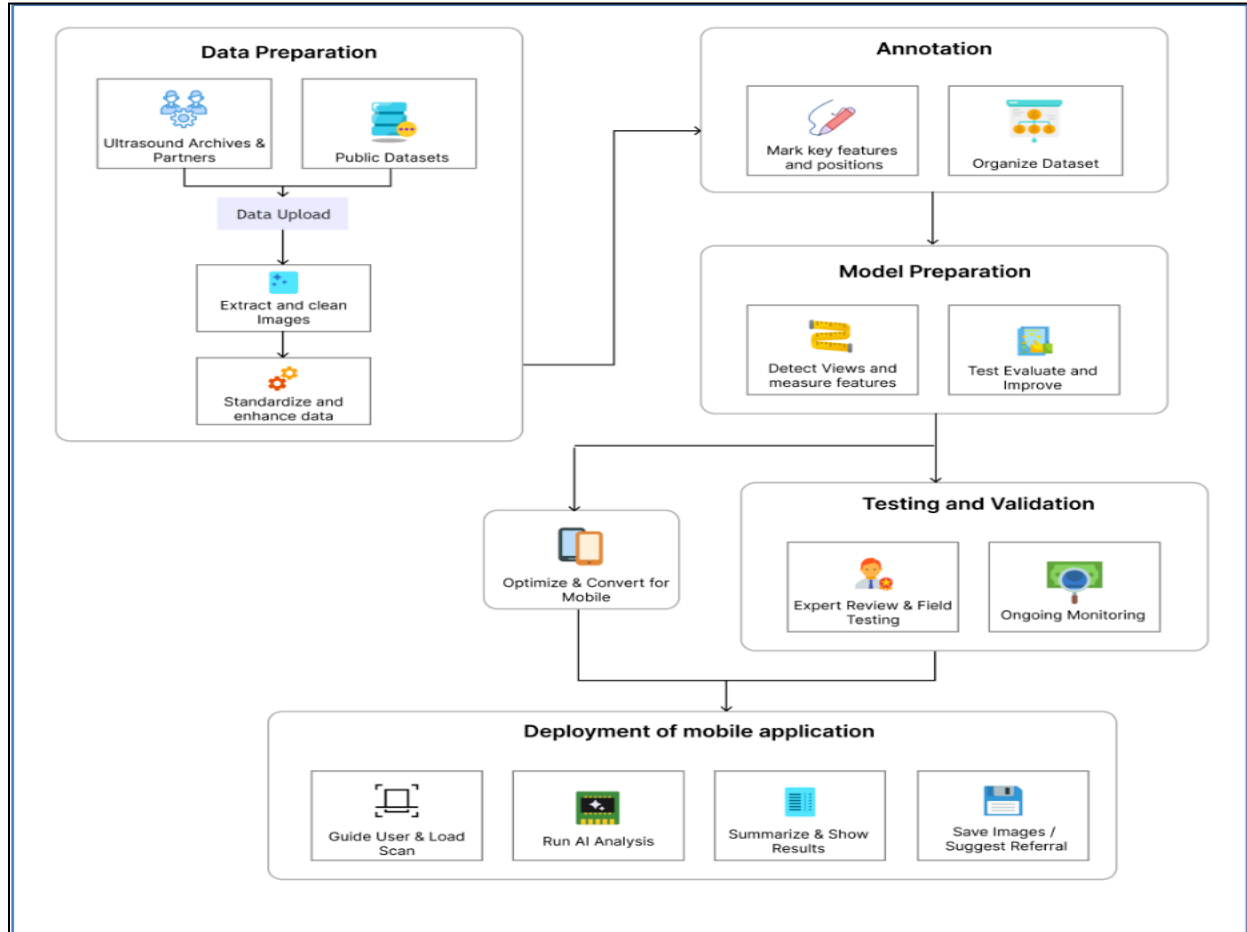


Fig. 1 Block Diagram of Natalis

The block diagram outlines the five-stage pipeline for developing and deploying the Natalis system. It starts with Data Preparation, where raw ultrasound images from archives and partners are cleaned, standardized, and enhanced for model training. In Annotation, experts label key fetal features and structures to build the ground truth dataset. During Model Preparation, specialized deep learning models are trained to detect views and measure biometric features, undergoing iterative testing to meet target accuracy. The Testing and Validation phase optimizes models for mobile use (e.g., TensorFlow Lite), followed by expert review and essential field testing by midwives. The final Deployment stage defines a streamlined user workflow: the app guides users through scans, runs offline AI analysis, summarizes results with risk flags, and saves data or suggests referrals—bringing AI-assisted prenatal diagnostics to low-resource settings.

4.2 Modular diagram representation of Natalis

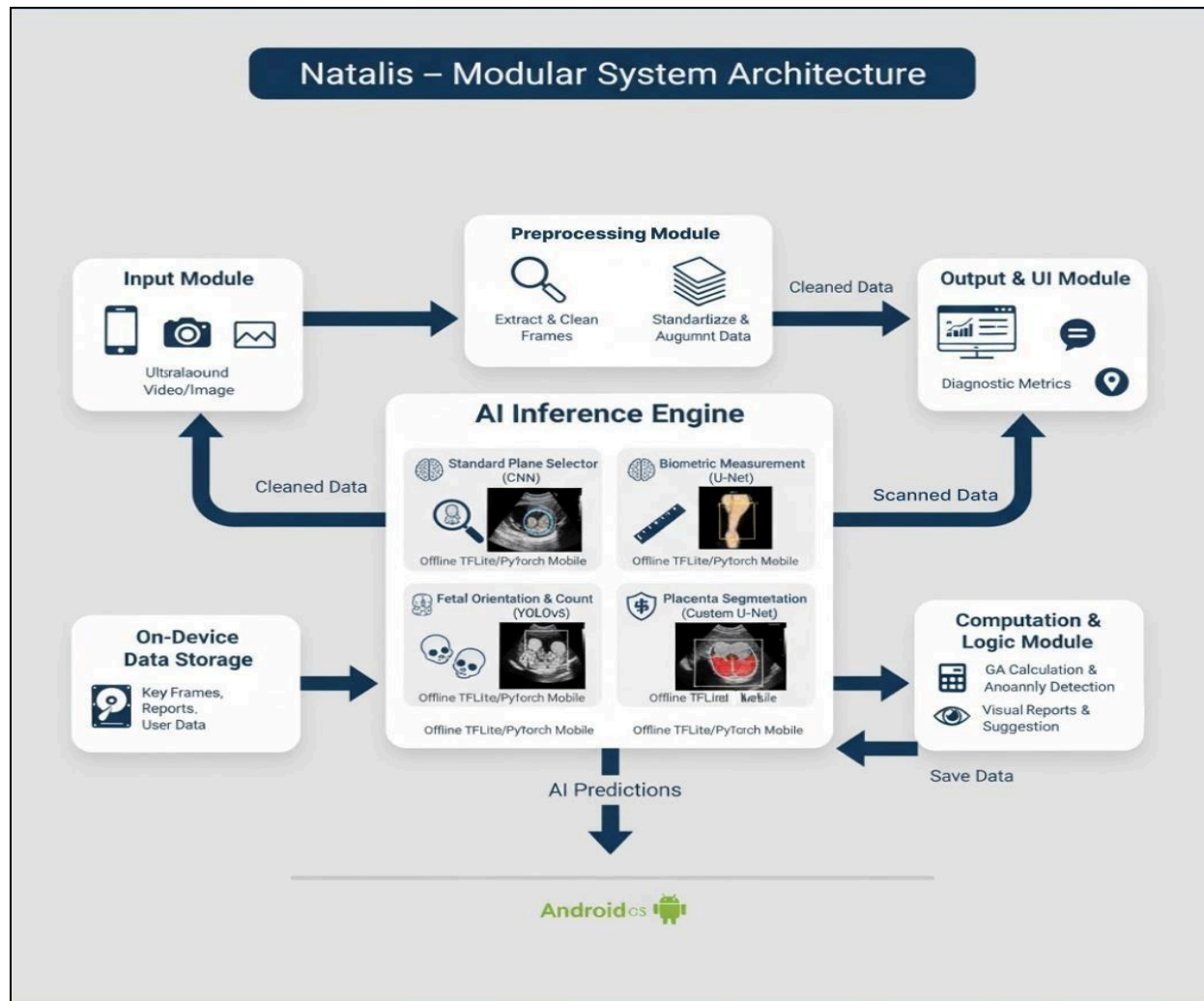


Fig. 2 Modular Diagram of Natalis

In the Computation & Logic Module, raw AI predictions are converted into actionable diagnostic metrics—calculating gestational age, confirming fetal count and presentation, and detecting anomalies like placenta previa. It also generates visual reports and referral suggestions. These outputs are sent to the Output & UI Module, which presents clear summaries and detailed reports via the Android interface. Simultaneously, the On-Device Data Storage module securely retains key frames, diagnostic reports, and user data locally, ensuring full offline functionality on Android OS.

4.3 Design of the proposed system

a. Data Flow diagram of Natalis

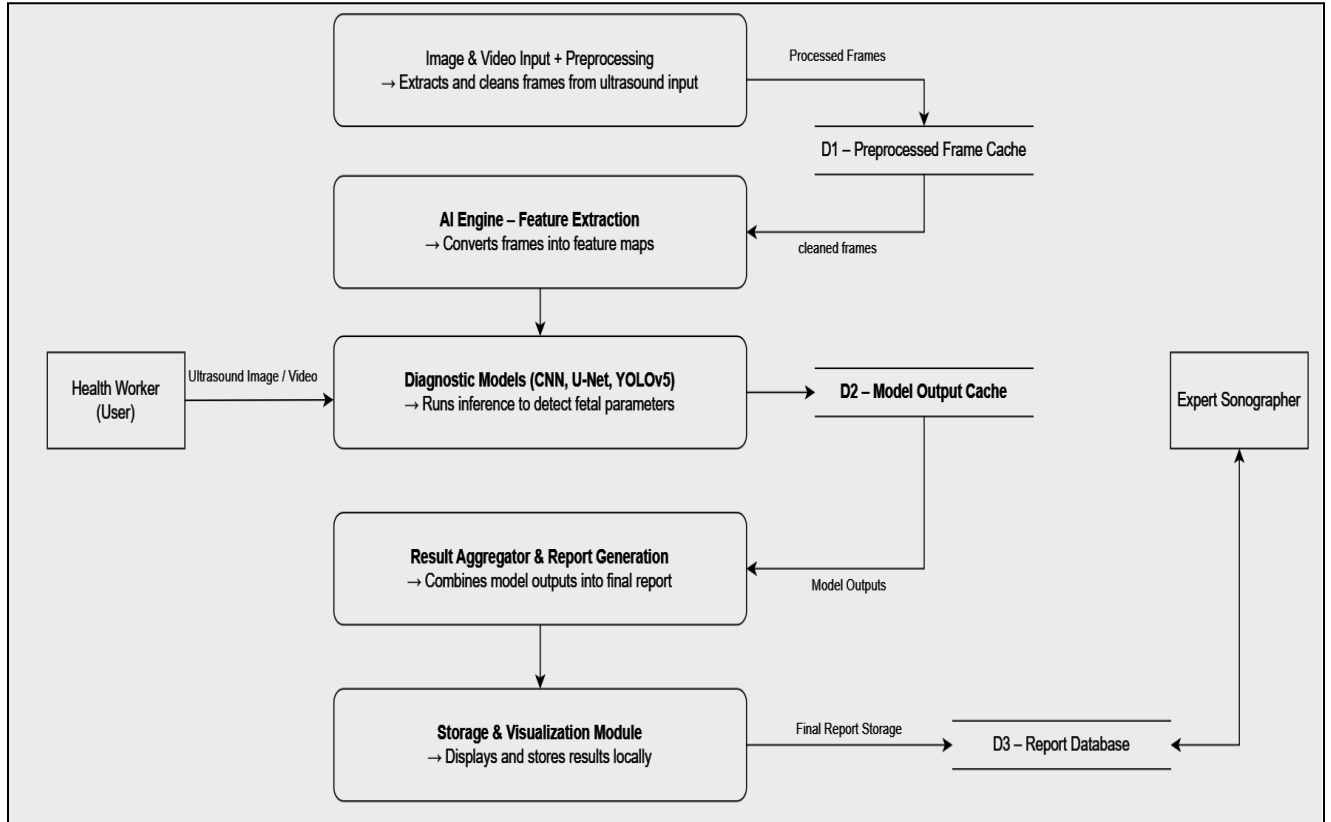


Fig.3 Data Flow Diagram of Natalis

The diagram illustrates a stepwise data flow where ultrasound images or video are input by the health worker, preprocessed, and feature-extracted via AI, then analyzed through diagnostic models. Results are aggregated into a report and stored locally, with expert sonographer input used for validation and cache databases supporting efficient processing at each stage.

b. ER diagram of Natalis

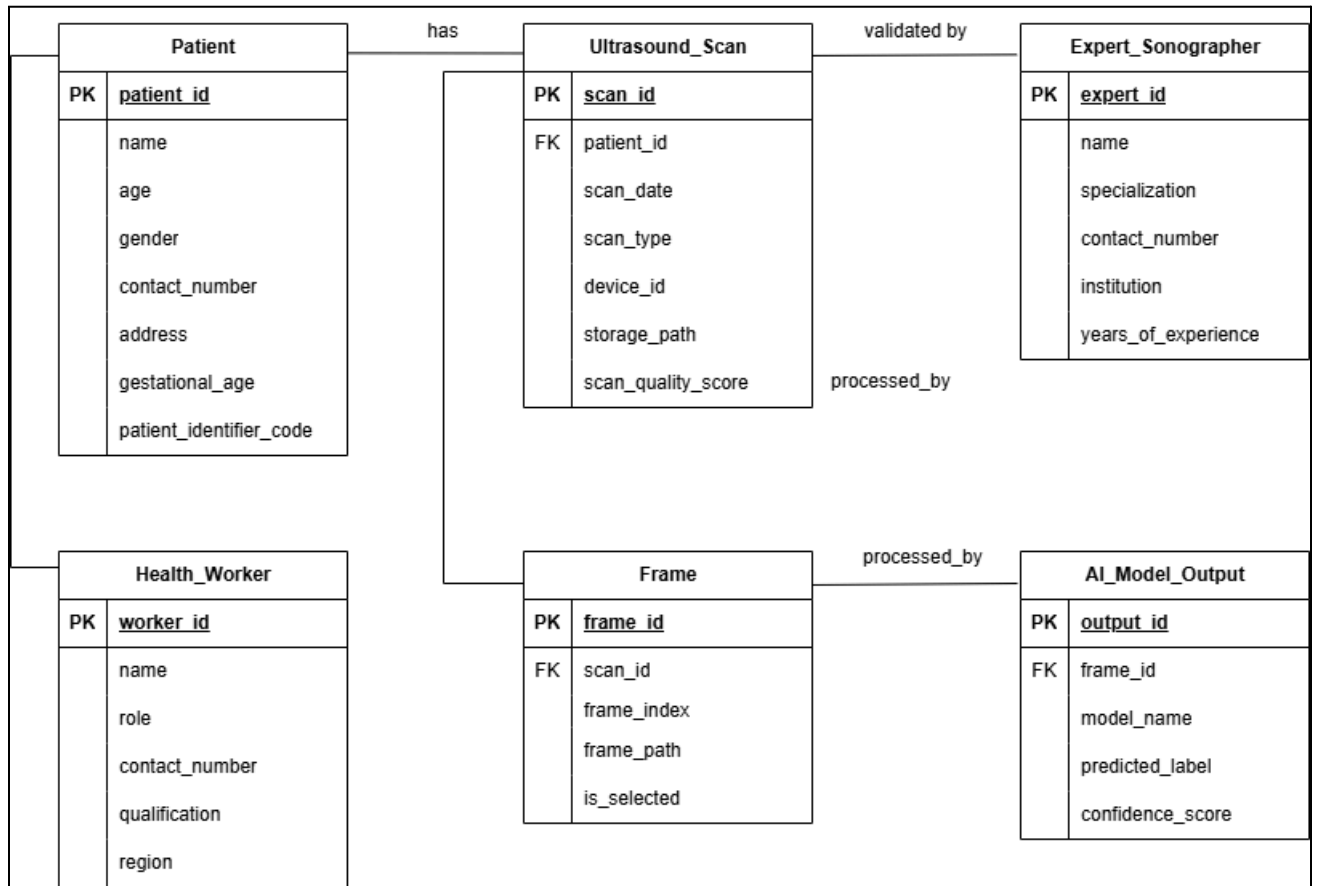


Fig. 4 ER Diagram of Natalis

This ER diagram models the database relationships for an AI-enabled ultrasound system, detailing entities like Patient, Ultrasound_Scan, Expert_Sonographer, Health_Worker, Frame, and AI_Model_Output. It shows how patient scans are linked to frames that are analyzed by AI and expert sonographers, with results and metadata stored for comprehensive healthcare recordkeeping.

c. Screenshots of the Implementation:

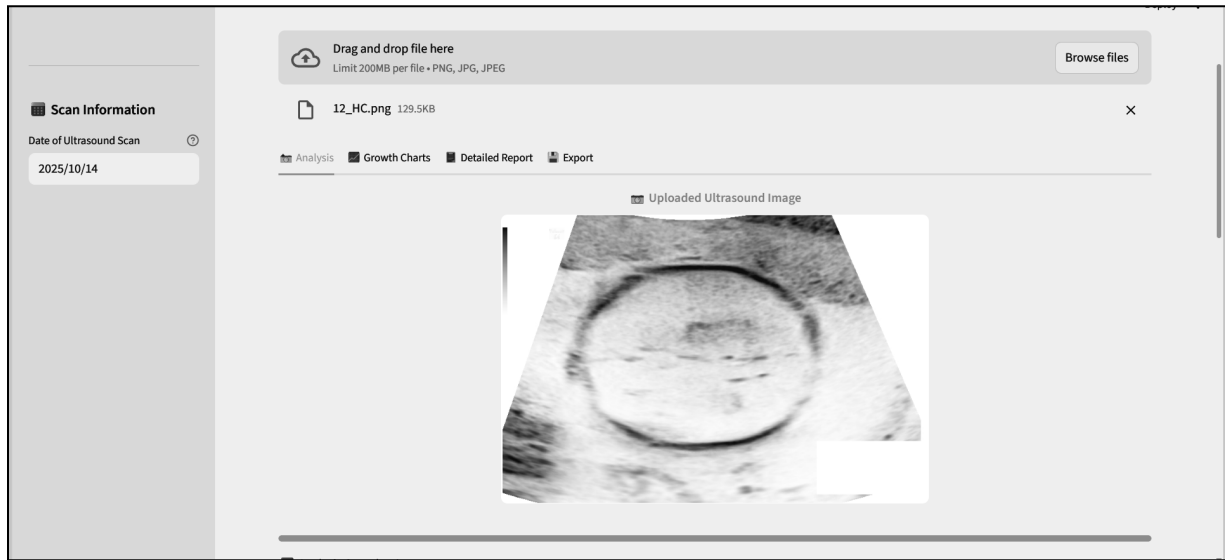


Fig. 5 Uploading Ultrasound image: Original fetal ultrasound image uploaded for analysis. Shows transventricular plane view of fetal head.

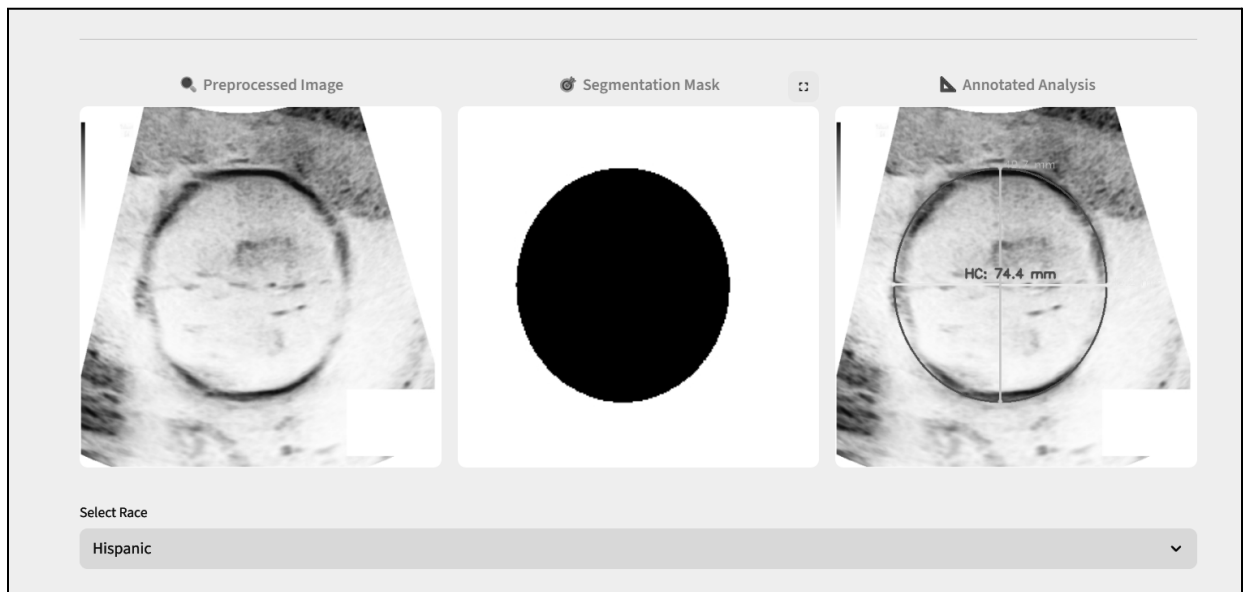


Fig. 6 Fetal Head Detection and Head Circumference: Fetal head segmentation with major/minor axes and calculated head circumference (HC) in millimeters.

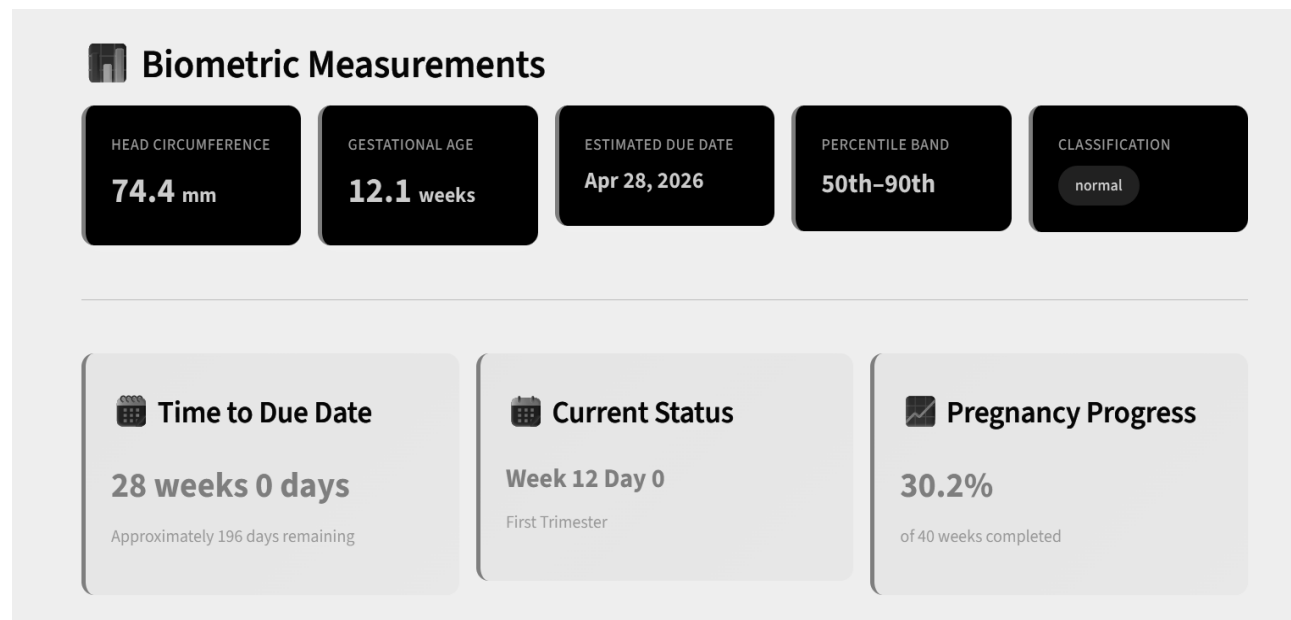


Fig. 7 Biometric measures : Comprehensive biometric analysis summary including HC, gestational age, estimated due date (EDD), and pregnancy progress indicators.

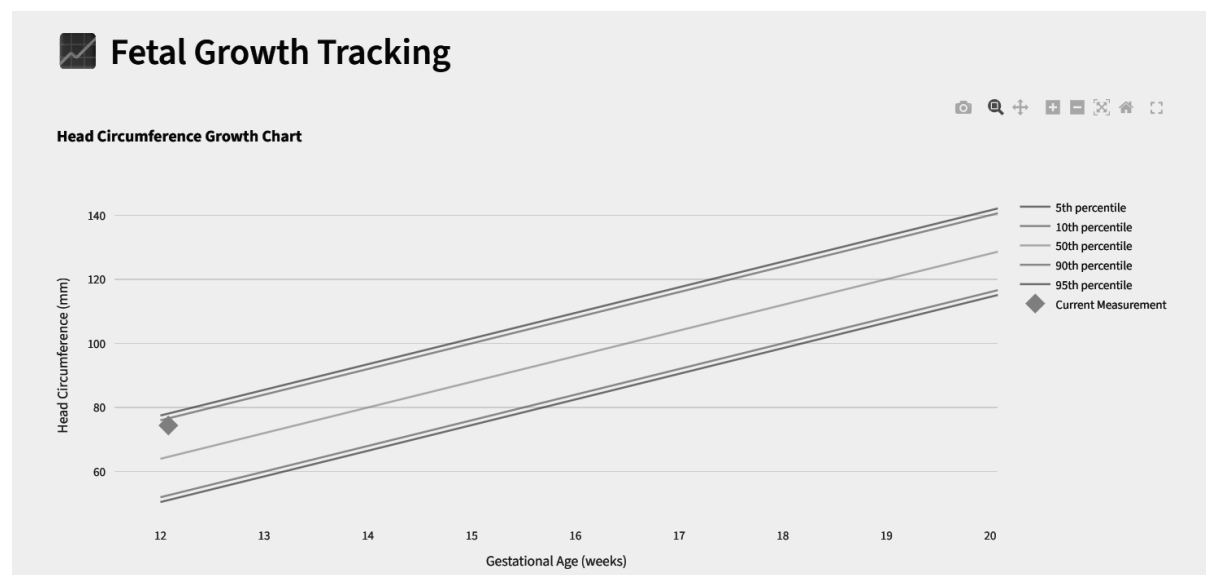
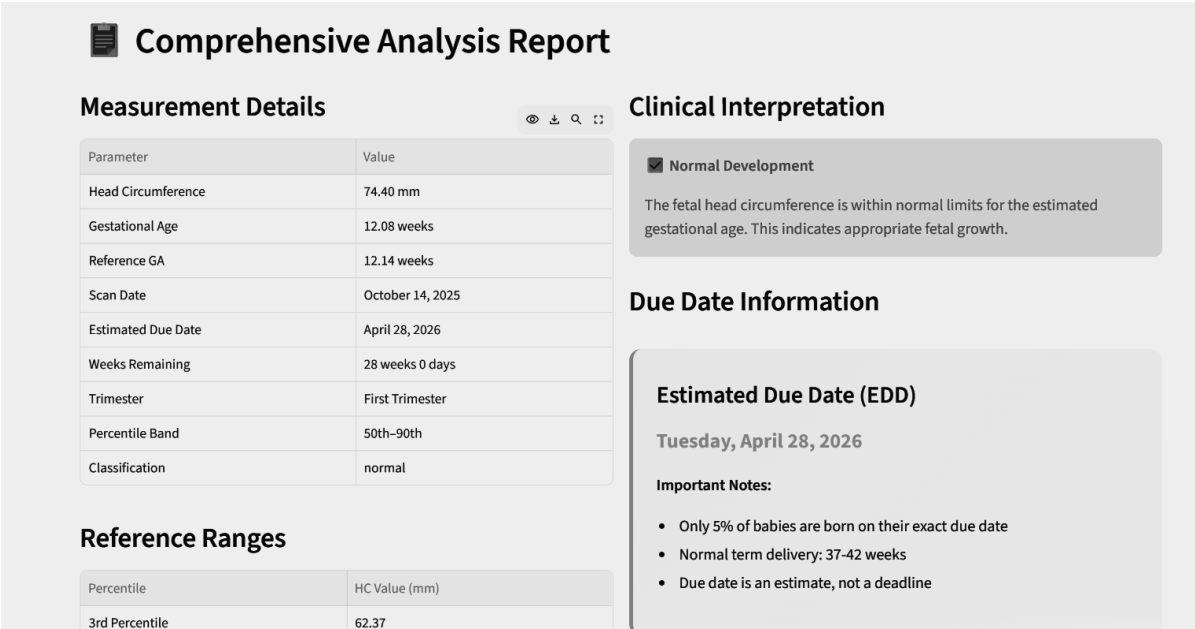


Fig. 8 Fetal Growth Graph Based on Gestational Age: Head circumference growth chart plotted against gestational age. Current measurement compared to population percentile curves



4.4 Algorithms Utilized in Existing Systems for Gestational Age Estimation

Table 4. Algorithms Utilized in Existing Systems along with their Application

Algorithm	How It Works	Inference / Application in GA Prediction
Convolutional Neural Networks (CNNs)	CNNs automatically extract spatial features from ultrasound images (such as fetal head, femur, or abdomen regions). They learn texture, shape, and edge patterns to predict gestational age directly from visual data.	Enables accurate and consistent GA estimation by analyzing fetal anatomical structures, reducing dependence on human measurement and image quality variations.
Attention-Based Hierarchical Neural Networks	Combines sequence learning and attention mechanisms to focus on important temporal or spatial patterns in Doppler or ultrasound signals. Learns fetal cardiac dynamics to infer developmental stages.	Provides highly accurate GA prediction using affordable handheld devices; useful in low-resource settings where image quality or trained operators are limited.
Support Vector Machine (SVM) Regression	Uses physiological signal features such as maternal and fetal heart rate variability (HRV) extracted from ECG or Doppler data. The algorithm finds an optimal hyperplane to predict GA as a continuous variable.	Enhances GA estimation by integrating physiological indicators; effective for small datasets and interpretable models, though less robust than deep learning for complex data.
Integral Projection with Morphological Operations	Preprocesses ultrasound images to remove noise, then applies projection-based segmentation to detect key fetal parts like the head or femur. Uses measurements such as Biparietal Diameter (BPD) or Femur Length (FL) for GA calculation.	Automates feature extraction and measurement, improving precision and consistency over manual methods. Suitable for portable ultrasound systems in clinical use.
Canny Edge Detection + Gaussian Filtering	Filters ultrasound images to smooth noise and applies edge detection to isolate fetal bone contours (especially femur). The detected contour length is used to estimate GA through regression or rule-based mapping.	Reduces observer variability and measurement errors; provides a simple, low-computation method suitable for real-time estimation on lightweight devices.

Proposed Results and Discussions

5.1.Determination of efficiency

Efficiency in Natalis refers to how quickly and resource-effectively the system processes ultrasound images or videos to produce diagnostic outputs.

Metrics:

1. Inference Time per Scan:

- Measure the average time taken for the AI engine to process one ultrasound video or a set of images.
- Expected target: ≤ 2 minutes per full scan on a mid-range Android device.

2. Resource Utilization:

- CPU/GPU load and RAM usage during on-device inference.
- Ensures compatibility with devices having ≥ 3 GB RAM and minimal battery drain.

3. Pipeline Throughput:

- Number of frames processed per second during video analysis.
- Evaluates how effectively frame extraction, preprocessing, and multi-task inference are optimized.

Evaluation Process:

- Run multiple ultrasound cases on the Android test device.
- Record processing times, RAM, and CPU usage.
- Compare against benchmark offline AI inference models to ensure optimal efficiency.

5.2 Determination of Accuracy

Accuracy assesses how closely Natalis’ predictions match expert sonographers’ evaluations.

Key Evaluation Tasks:

Table 5. Key Metrics

Task	Metric	Target
Gestational Age (GA)	Mean Absolute Error (MAE)	$\pm 1\text{--}2$ weeks
Fetal Presentation	Classification Accuracy	$>95\%$
Placental Segmentation	Dice Similarity Coefficient	≥ 0.90
Fetal Count Detection	True Positive Rate / F1 Score	$>95\%$

Procedure:

1. An annotated ultrasound dataset used as ground truth.
2. Predictions compared against expert annotations.
3. Compute **MAE, precision, recall, F1-score, and DSC** for each diagnostic task.
4. Aggregate results to evaluate overall system accuracy.

5.3 Reports on Sensitivity Analysis

Sensitivity analysis evaluates how **robust Natalis is to variations** in input data, preprocessing parameters, or model hyperparameters.

Scenarios Tested:

1. **Image Quality Variations:** Low-resolution, noisy, or partially occluded ultrasound frames.
2. **Angle and Orientation Variations:** Slightly tilted or rotated fetal views.
3. **Model Parameter Changes:** Different learning rates, batch sizes, or augmentation settings.

Outcome Measures:

- **Effect on GA estimation:** Change in MAE under varying input conditions.
- **Detection robustness:** Drop in fetal presentation or count accuracy.
- **Segmentation stability:** Variation in the Dice coefficient for placental detection.

Reporting:

- Tables summarizing **performance drop (%)** for each scenario.
- Identify which conditions the system is most sensitive to, guiding dataset augmentation and model refinement.

5.4 Graphs: Accuracy vs Time

Visualizing how system accuracy changes with **processing time, dataset size, or training iterations** helps assess performance trade-offs.

Types of Graphs:

1. **Inference Accuracy vs Processing Time (On-device)**
 - X-axis: Time taken per scan (seconds)
 - Y-axis: Accuracy (%) for GA, presentation, or fetal count
 - Purpose: Determine if faster processing sacrifices diagnostic accuracy.
2. **Training Epochs vs Accuracy**
 - X-axis: Epochs
 - Y-axis: Validation accuracy or Dice coefficient
 - Purpose: Show model convergence and optimal stopping point.
3. **Dataset Size vs Accuracy**
 - X-axis: Number of annotated images/videos
 - Y-axis: Accuracy (%)
 - Purpose: Demonstrates the impact of dataset expansion on model performance.

Visualization Tools:

- Matplotlib, Seaborn for Python training graphs.
- TensorBoard for real-time monitoring of model metrics

Visualization Plots:

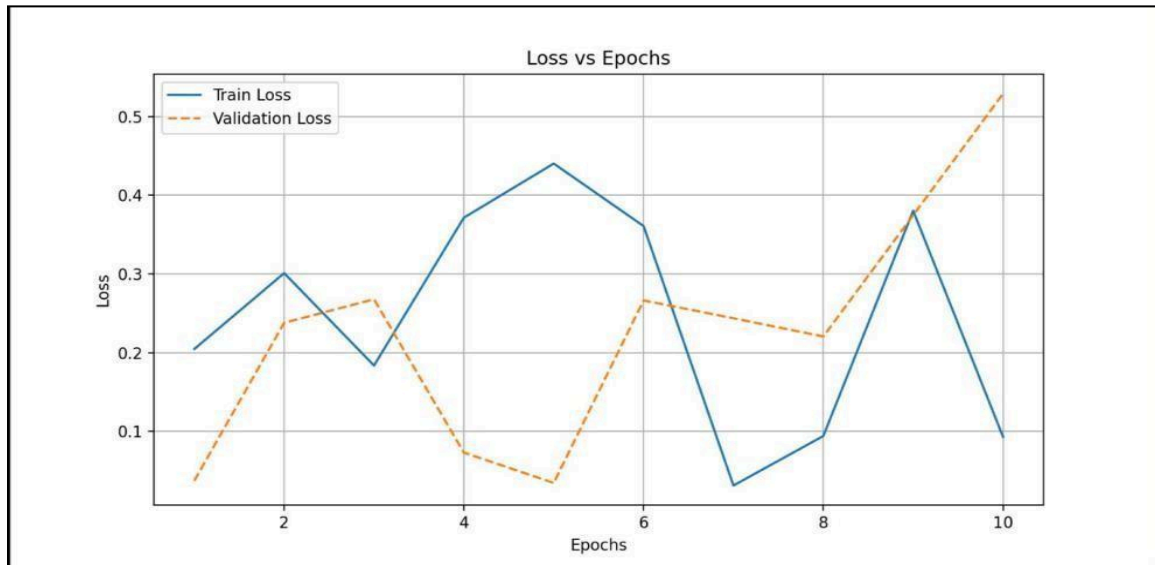


Fig.11 Loss vs Epochs graph

The Validation Loss (dashed line) shows a sharp rise after epoch 8, indicating the model is likely overfitting to the training data. The minimum validation loss was achieved around epoch 5.

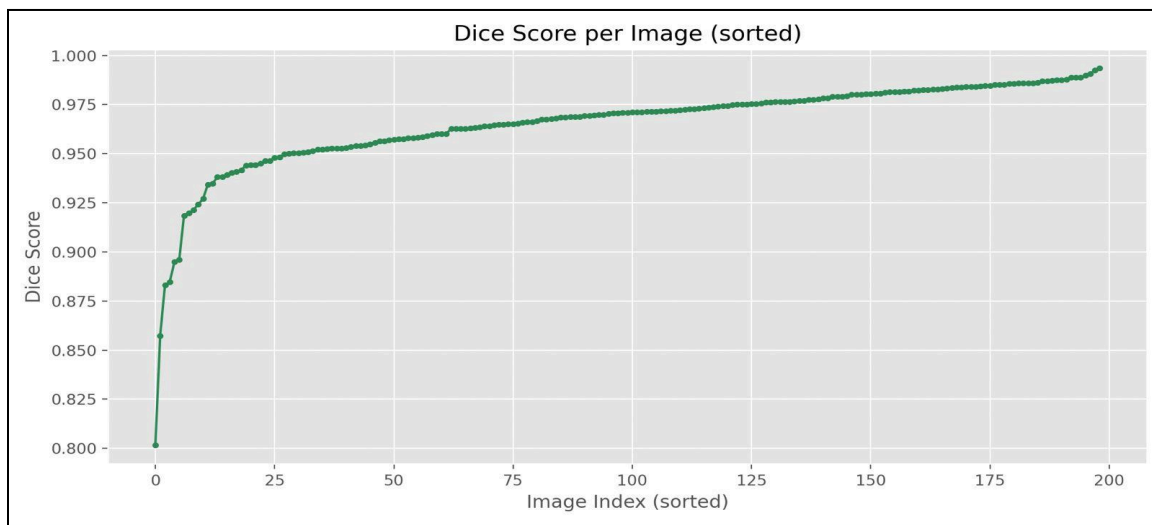


Fig.12 Dice Score per Image

This plot shows scores rapidly increasing from a minimum of 0.80 for a few images, quickly rising, and then stabilizing at a high level. Only a small fraction of images had poor segmentation scores.

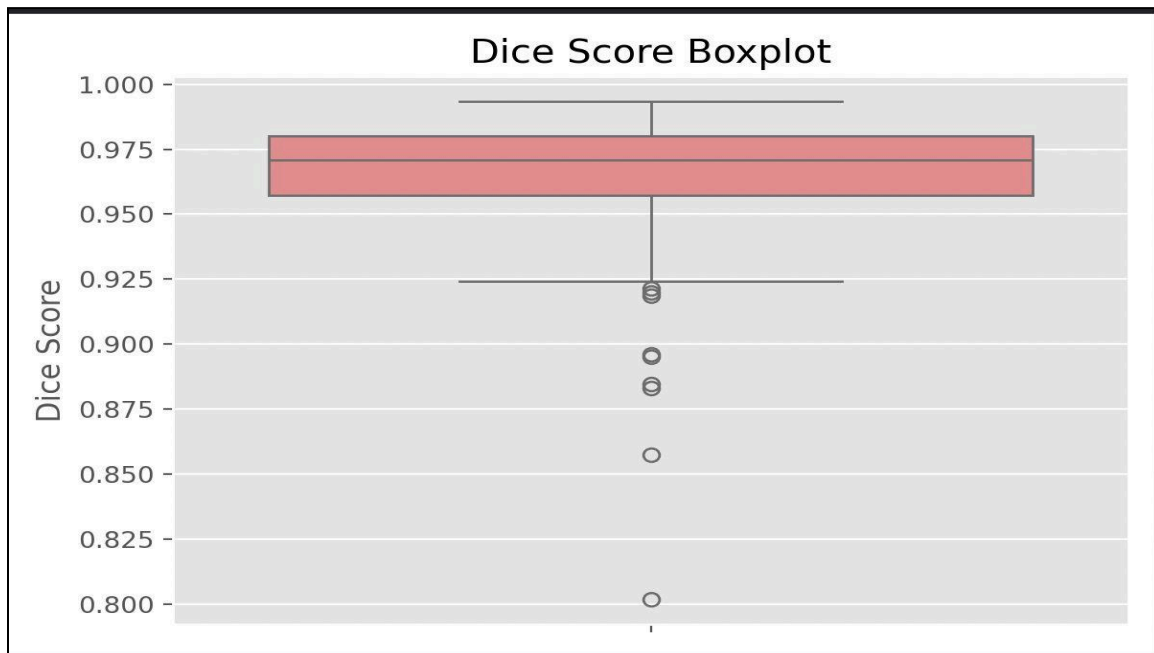


Fig.13. Dice Score Boxplot

The majority of Dice Scores are very high, concentrated between 0.96 (Q1) and 0.975 (Q3), with a high median around 0.97. A few significant outliers show lower performance, down to 0.80

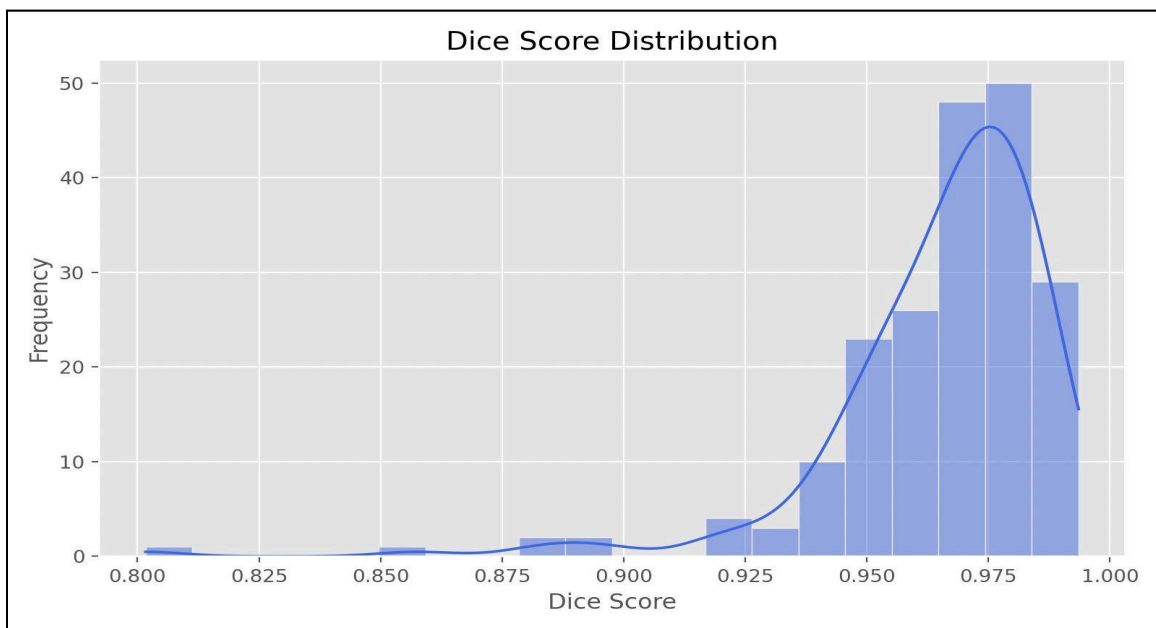


Fig.14 Dice Score Distribution

The distribution of scores is heavily left-skewed, with the highest frequency of images scoring in the 0.975 to 1.000 range. This confirms the model achieved high accuracy on most images.

Plan Of Action For the Next Semester

6.1 Work Done Till Date

Basic Model Development

- Implemented a **deep learning model for fetal head detection**.
- Model tested on sample ultrasound images using **Streamlit** for rapid prototyping and visualization.
- Verified preliminary accuracy of head detection for gestational age estimation.

Prototype Interface

- Developed a **Streamlit-based interface** to:
 - Upload ultrasound images.
 - Display detected head region.
 - Provide basic GA estimation.
- Served as a **proof-of-concept** for offline AI inference on images.

Model Training & Preprocessing

- Curated a small dataset for head detection.
- Preprocessed images (resizing, normalization, augmentation) to improve model robustness.

6.2 Plan of Action for Project

The next semester will focus on **scaling the prototype into a full-featured mobile application** with a complete ML pipeline design and additional diagnostic capabilities:

Full Mobile Application Development

- Transition from **Streamlit prototype** to an **Android-based app**.
- Integrate AI models for:
 - Gestational age estimation (head, femur, abdomen)
 - Fetal presentation detection (cephalic/breech)
 - Placenta segmentation
 - Fetal count detection (twins, triplets)

Design a **user-friendly interface** with:

- Multilingual support
- Visual reports (icons, color-coded alerts)
- Offline functionality

System Design of ML Models

- Define a **modular ML pipeline** for the mobile app:
 - Frame extraction → Preprocessing → Sequential inference (CNN, U-Net, YOLOv5) → Report generation.
- Optimize models for **on-device efficiency**:
 - TensorFlow Lite / PyTorch Mobile conversion
 - Quantization and pruning for reduced memory and faster inference
- Ensure a **scalable architecture** to add new features or models later.

Dataset Expansion & Annotation

- Collect and label images/videos for:
 - Standard planes (head, femur, abdomen)
 - Placenta positioning
 - Fetal orientation and count
- Include **diverse patient profiles and imaging conditions** to improve model robustness.

Testing & Evaluation

- Evaluate the **accuracy, efficiency, and usability** of the mobile system.
- Conduct **pilot studies** with frontline health workers to compare AI outputs against expert sonographers.
- Perform **sensitivity analysis** to assess model robustness under variations in input quality.

Documentation & Reporting

- Prepare a **detailed system design document** of the ML architecture.
- Document evaluation metrics, inference time, and UI/UX feedback.
- Generate **accuracy vs time graphs** and sensitivity analysis reports for submission.

Conclusion

Natalis aims to democratize prenatal screening by bringing AI-powered ultrasound capabilities to underserved and low-resource areas. By working entirely from existing ultrasound images and videos, and running completely offline on a smartphone, Natalis removes common barriers such as unreliable electricity, limited internet connectivity, and the shortage of highly trained sonographers. The system's AI core will provide essential obstetric insights, including gestational age estimation, fetal position, and placental health assessment, directly at the point of care. By guiding health workers through simple steps and delivering clear, actionable results, Natalis will enable the timely identification of high-risk pregnancies that require referral to advanced care. If successfully implemented, Natalis has the potential to significantly improve antenatal care coverage in areas where access to ultrasound is currently limited. More women will be able to receive the WHO-recommended mid-pregnancy scan, and preventable complications could be detected earlier, improving outcomes for both mothers and babies. This project plan sets out a clear pathway from data preparation and AI algorithm development to the deployment of a fully functional mobile application, laying the foundation for a cost-effective, scalable, and life-saving maternal health solution

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Appendix

a. List of Figures

Figure Number	Heading	Page no.
1	Block Diagram of Natalis	14
2	Modular Diagram of Natalis	15
3	Data Flow Diagram of Natalis	16
4	ER Diagram of Natalis	17
5	Uploading Ultrasound image	18
6	Fetal Head Detection and Head Circumference	18
7	Biometric measures	19
8	Fetal Growth Graph Based on Gestational Age	19
9	Fetal Measurements and Estimated Due Date	20
10	Head circumference reference ranges	20
11	Loss vs Epochs Graph	26
12	Dice Score Per Image	26
13	Dice Score Boxplot	27
14	Dice Score Distribution	27

b. List of tables

Table Number	Heading	Page no.
1	Literature Survey of Research Papers	6
2	Hardware and Software Requirements by Categories	12
3	Tools and Libraries used	13
4	Algorithms Utilized in Existing Systems along with their Application	21
5	Key Metrics	24